

Advanced Python OOP Notes - CS101

Class Variables vs Instance Variables Instance variables are unique to each instance (like a specific employee's name). Class variables are shared among all instances of a class (like an annual company-wide raise amount).

The Danger of Attribute Shadowing If you attempt to update a shared class variable by using an instance (for example, typing `employee_1.raise_amount = 1.05`), Python will not change the class variable. Instead, it creates a brand new instance variable just for `employee_1` that hides the class variable. This specific behavior is known in computer science as "Attribute Shadowing," and it is a common source of bugs for junior developers.